

420-D03-AB — Cyber Defense

Final Project & Course Procedure Manual — Student Brief

AEC Cybersecurity LEA.D8 · Cohort CYB-01 · Cyber.SoHo Educational Hub

Worth	Final Project 15% + Course Procedure Manual 10% = 25% of your final grade
Built from	The six hands-on labs you have already done (a further 30%)
Submission deadline	Wednesday 21 October 2026, 23:59 EDT (midnight)
Project work sessions	Class 18 (Oct 13) · 19 (Oct 14) · 20 (Oct 17) · 21 (Oct 20)
Final Review	Class 22 — Wednesday 21 October
Final Exam	Class 23 — Saturday 24 October
Competency	EZ03 — To apply access control measures (elements 1–4)

One deadline to remember. Everything is due at **midnight on 21 October, Eastern Daylight Time**. The Final Review that same evening is your last chance to ask questions — not your chance to start. Late submission carries a **10% deduction per calendar day** under departmental policy.

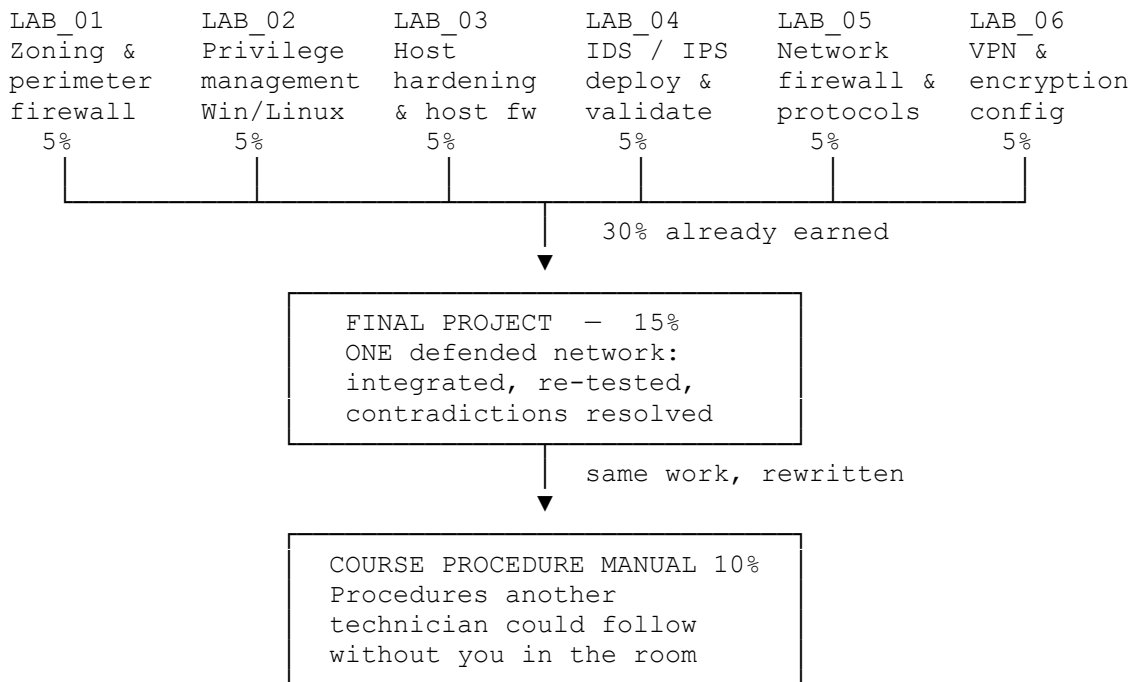
1. The short version

You are not starting a new assignment. **You are assembling one you have been building since 29 August.**

Every lab produced four things: a working configuration, screenshot evidence, a test result, and a short written finding. Six labs × four artifacts = **24 pieces already in your hands**. The Final Project is those pieces made into one coherent, defended network. The Course Procedure Manual is the same work written so somebody else could rebuild it without you.

If you kept up with the labs, this is two evenings of assembly and editing. If you did not, this is the week you find out why I said *"write the lab up the same evening you do it"* on day one.

2. THE WRITTEN PROJECT — HOW THE WHOLE THING FITS TOGETHER



$30\% + 15\% + 10\% = 55\%$ of your grade, ONE body of work

Read the diagram in one sentence: the labs are the parts, the Project is the machine, and the Manual is the instruction book for the machine.

The three jobs, and how they differ

	The six labs (done)	The Final Project (15%)	The Manual (10%)
Question it answers	<i>Can you configure this control?</i>	<i>Do your controls work together as one design?</i>	<i>Could someone else reproduce your result?</i>
Audience	Your instructor	A technical reviewer / client	A technician who has never seen your network
Voice	"I configured..."	"The network is designed so that..."	"Step 1. Do this. Expected result: ..."
Fails when	The control does not work	The six labs contradict each other	It cannot be followed without asking you a question

Where the contradictions live — check these first

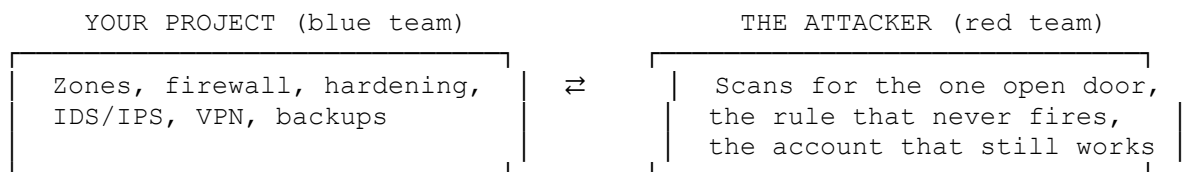
When six labs get merged, the same four collisions appear every single year:

1. **Address overlap.** LAB_01 zoning and LAB_05 firewall rules disagree about which subnet a zone actually uses.
2. **Rule order.** A permissive rule written in LAB_01 sits above a deny rule written in LAB_05, so the deny never fires.
3. **Account drift.** The service account created in LAB_02 is not the one the LAB_04 sensor or the LAB_06 VPN actually authenticates with.
4. **Protocol contradiction.** LAB_03 disabled a legacy protocol that a LAB_05 or LAB_06 configuration still depends on.

Finding and resolving these is the Final Project. A submission that lists three contradictions it found and explains how it fixed them scores higher than one that pretends there were none.

2b. How your defence gets tested — the attacker's-eye view

Here is the thing nobody tells you until later: **the network you are defending in this project is exactly the kind of network an intrusion tester is paid to break into.** In the Intrusion Testing course, another cohort builds an attack lab and points it at configurations that look just like yours. So the smartest way to check your own work is to stop thinking like a builder for an hour and think like the attacker.



The four "contradiction collisions" above are the doors the attacker walks through.

Every contradiction you leave in is a door. The permissive rule sitting above a deny rule (collision 2) is not a paperwork problem — it is the exact gap a tester's `nmap` scan lights up. The service account that still works after you thought you locked it down (collision 3) is how they move sideways. When you hunt these down for your integration findings, you are doing the attacker's reconnaissance *for* them, and then closing the door before they arrive.

What this means for your marks. Section 5 (verification) is no longer just "I turned the control on." The strong version is: **"I attacked my own network and proved the control caught it."** See the upgraded verification requirement below.

Prove your defence detects a real attack (not just that it is configured)

This is legitimate, in-scope defensive work — the competency literally requires *"rigorous validation of the effectiveness of the access control measures."* On your own lab, against your own machines only:

1. From a Kali VM (or any host you control on the lab network), run an authorised `nmap` scan and **one** controlled test against a target you own — for example an Atomic Red Team test that your IDS/IPS should catch.
2. On the defender side, **capture it** — Wireshark on the LAN, plus your Suricata/Zeek/Wazuh alert if you deployed one.
3. Write it up: what you fired, what the defence saw, and — honestly — what it *missed*. A control that blocked silently and logged nothing is half a control.

That write-up is worth more than any screenshot of a settings page, because it is evidence the defence **works**, not just evidence it **exists**.

3. What to submit — Part A: the Final Project

One document. Aim for **10–16 pages**, excluding evidence appendices.

§	Section	What it must contain
1	Executive summary	One page a non-technical partner could read. What the network protects, what it now resists, what remains a known risk.
2	Reference network	The zone diagram, the address plan, the device inventory. One picture, one table.
3	Defence-in-depth map	Each of the seven layers, the control you built at it, and the lab it came from.
4	Integration findings	The contradictions you found merging the six labs, and how you resolved each. Minimum three.
5	End-to-end verification	Tests run on the <i>assembled</i> network — not the per-lab tests. Each with expected vs. actual result. Include at least one adversarial test: attack your own network and prove a control detected or blocked it (see §2b).
6	Residual risk	Use the layer model. State your residual figure and your correlated failure modes. An analysis that hides its assumptions is not an analysis.
7	Recommendations	What you would do next with more time or money, in priority order.
8	Evidence appendix	Screenshots, keyed to the evidence register. Every one carries an ID and a claim.

4. What to submit — Part B: the Course Procedure Manual

The same work, re-expressed as procedures. **One chapter per lab, seven sections each** — exactly the skeleton `rle01_cpm_scaffold.py` generated for you on day one.

§	Section	The test it must pass
1	Objective	Names the competency element it satisfies
2	Scope and assumptions	States what it covers and what it does not
3	Pre-conditions	Snapshots, credentials, change window
4	Procedure	Numbered. One action per step. No step needs you to explain it
5	Verification	The test that proves the control works, with the expected result
6	Rollback	How to undo it when the change breaks production
7	Evidence	File names, command output, capture dates — cross-referenced

Plus a cover page with a **document-control table** (version, date, author, change) and the **evidence register CSV**.

The Manual test: hand it to a classmate who was not in your group. If they can follow any chapter start to finish without asking you a single question, it is finished. If they ask even one, that answer belongs in the document.

4b. Where you build all this — lab environment

If you are assembling the project on your **own laptop** (the common case), here is what you need. Everything runs offline in Oracle VirtualBox — nothing bridges to the real Internet, and your only targets are your own VMs.

Component	Minimum (works)	Recommended (comfortable)
CPU	4 cores, virtualisation (Intel VT-x / AMD-V) enabled in BIOS/UEFI	6–8 cores
RAM	16 GB	32 GB — the single biggest comfort factor
Free disk	120 GB free (SSD)	200 GB on NVMe SSD
Host OS	Windows 10/11 64-bit, macOS 12+ (Intel), or Linux	Same, fully updated

Software: Oracle VirtualBox 7.1.x + the matching Extension Pack (free for educational use); an OPNsense or pfSense CE ISO for the edge firewall; a Windows evaluation VM and a Debian/Ubuntu Server VM as the hosts you defend; Wireshark; and — for the adversarial verification in §2b — a light Kali VM.

The #1 support ticket, every term: if VirtualBox only offers 32-bit guests or a VM won't start with a `VT-x` error, virtualisation is switched **off** in your firmware. Reboot into BIOS/UEFI and enable **Intel VT-x** or **AMD-V / SVM**. On Windows, also disable Hyper-V / "Memory Integrity" if it conflicts. Do this **before** the work session, not during it.

A note on LAB_06 (VPN): the modern choice is **WireGuard**, which is now built into OPNsense 25.7 in-core — no plugin needed. It is faster than OpenVPN with a smaller attack surface. If your LAB_06 used OpenVPN, mention in your write-up why WireGuard is the current best practice.

5. Your calendar from here

Class	Date	What happens
16–17	Oct 7 & 10	LAB_06 — VPN setup and encryption config. Last lab.
18	Tue Oct 13	Q_04 + project work begins. Bring all six labs assembled into one folder.
19	Wed Oct 14	Project work — integration findings and end-to-end testing.
20	Sat Oct 17	Project work — the long session. Manual writing.
21	Tue Oct 20	Project work — final editing and evidence check.
22	Wed Oct 21	Final Review — and your 23:59 EDT submission deadline .
23	Sat Oct 24	Final Exam (25%) .

If you saw an earlier draft naming Class 16 as the due date, this brief supersedes it. The labs finish 10 October; the project is built across Classes 18–21 and is due at midnight on 21 October.

6. How it is marked

Criterion	Weight of the 25%	You lose marks when
Integration — is it one network, not six exercises?	25%	The six labs are pasted together unedited
Verification — did you test the assembled system?	20%	You reused per-lab test results
Documentation quality — could a stranger follow it?	20%	Steps assume knowledge only you have
Evidence — is every claim supported?	15%	Screenshots with no claim attached
Risk reasoning — residual risk + correlated failures	10%	Assumptions are hidden or unstated
Professional presentation — versioning, structure, clarity	10%	No version table, no page numbers, no rollback

Declare any AI assistance. Tool, model, and what you verified yourself. Undeclared AI-written text is an academic-integrity matter under the College's *Academic Integrity Cheating & Plagiarism Procedure and Policy No. 7 (IPESA)*. Declared and verified use is fine — it is the professional norm now.

The five ways students actually lose marks — and the fix

The trap	The fix
Six labs pasted together, still contradicting each other	Do §2b — hunt the four collisions, resolve them, write it up
"I configured the firewall" with a screenshot of the settings page	Prove it <i>works</i> : an adversarial test that the control caught (§2b)
Evidence with no claim — a folder of unlabelled screenshots	Every evidence row names what it proves , not just a file name
A residual-risk number with no assumptions	State the figure and your correlated failure modes
A Manual only you could follow	The classmate test — one question means one missing step

7. Submission checklist

Tick all of these before you upload:

- ☐ Final Project document, sections 1–8 complete
- ☐ Course Procedure Manual, six chapters × seven sections
- ☐ Cover page with document-control table and your student number
- ☐ `EVIDENCE_REGISTER.csv` — every row has a **claim**, not just a file name
- ☐ Minimum three integration findings, each with a resolution
- ☐ End-to-end verification results from the **assembled** network
- ☐ At least one **adversarial test** — you attacked your own network and proved a control caught it (§2b)
- ☐ Residual-risk figure **and** correlated failure modes stated
- ☐ No passwords, keys or client data anywhere in the documents
- ☐ AI assistance declared, if used
- ☐ Single archive named `CYB01_LastName_FirstName_FinalProject.zip`

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